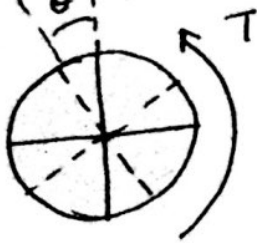


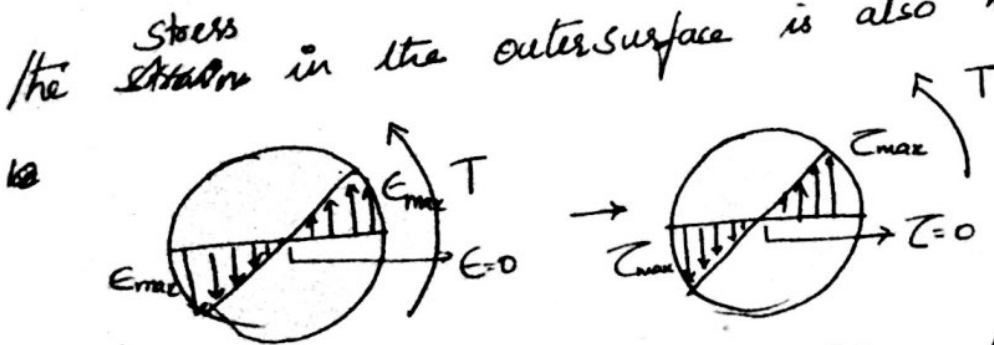
Torsion in Circular Bars/Cylinders:

Machine parts are loaded in torsion, either to transmit power or support a dynamic load.

When we apply a torque ' T ' to a circular shaft, the shaft will twist through an angle ' θ '. When the material twists the specimen undergoes deformation and hence we have strain.



When the specimen is twisted by an angle θ , we can observe that the strain in the outer surface is max and the stress in the outer surface is also maximum.



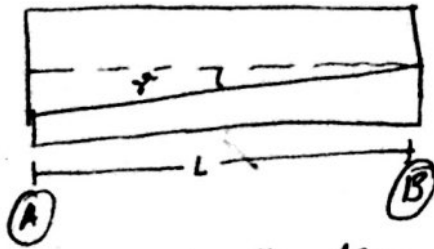
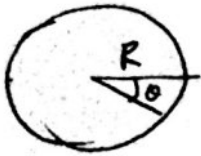
Hence, $\epsilon=0$, and $\tau=0$ at the center of the circular shaft. and ϵ_{max} and τ_{max} at the outermost surface.

Assumptions of Torsion Theory:-

- ① Material is homogeneous,
- ② Follows Hooke's Law, i.e. elastic
- ③ No warping, i.e. only angular displacements.

Torsion Equation:-

(2)



When the end (A) is twisted, the line rotates through an angle θ and the length of the arc produced is $R\theta$. and in the longitudinal direction, we have angle ϕ forming with respect to the axis.

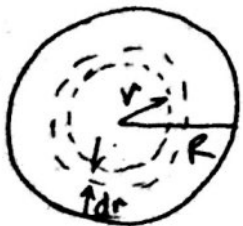
Hence, the length of the arc produced is $L\phi$.

$$\therefore L\phi = R\theta \text{ and } \phi = \frac{R\theta}{L} \quad \text{--- (1)}$$

From hooke's law for shear stress, shear strain relation.

$$G = \frac{\tau}{\phi} \quad \text{--- (2)}, \text{ substituting (1) in (2)}$$

$$\frac{R\theta}{L} = \frac{\tau}{G} \quad \text{or} \quad \frac{G\theta}{L} = \frac{\tau}{R} \quad \text{or} \quad \tau = \frac{G\theta R}{L} \quad \text{--- (3)}$$



$$\text{Area of the circle} = \pi R^2 = A$$

$$\frac{dA}{dr} = 2\pi R \rightarrow$$

Shear force on the elemental ring is $\frac{dF}{dA} = \tau$

$$dF = \tau \cdot dA = \tau \cdot 2\pi r \cdot dr$$

The torque applied is, $T = F \cdot r$

$$dT = \tau \cdot 2\pi r \cdot dr \cdot r = \tau \cdot 2\pi r^2 \cdot dr \quad \text{--- (4)}$$

Equating (3) and (4), we get

$$dT = \frac{G\theta 2\pi r^3 \cdot dr}{L}$$

Integrating we get.

(3)

$$T = \frac{G\theta 2\pi}{L} \int_0^R r^3 dr$$

The term $2\pi \int_0^R r^3 dr$ is called polar moment of ~~Inertia~~ Inertia, or polar second moment of area.

and denoted as 'J'.

$$\text{Hence, } T = \frac{G\theta J}{L} \quad \text{--- (5)}$$

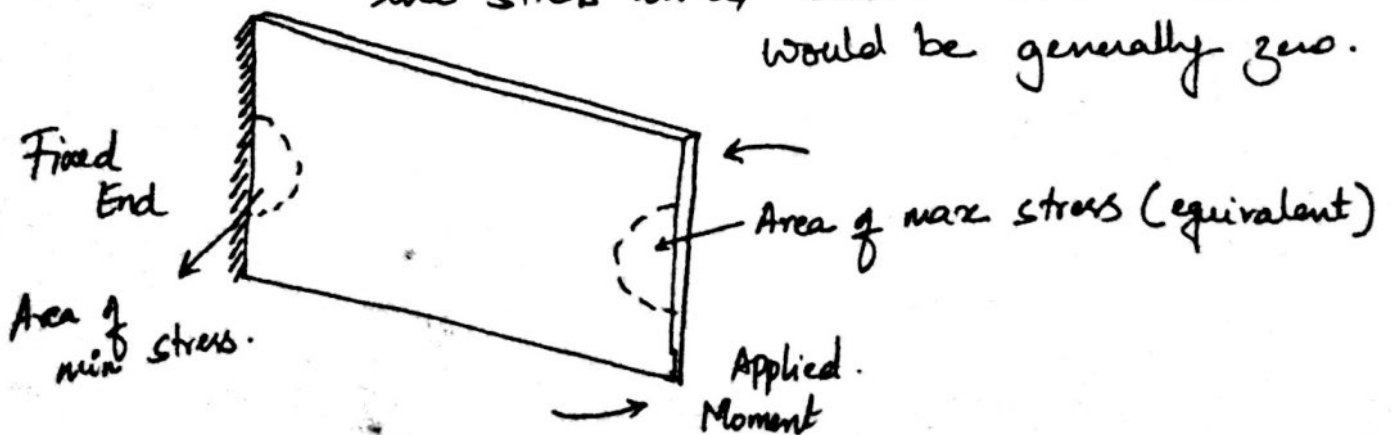
Combining (3), (4) and (5)

$$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R} \rightarrow \text{Bending Equation}$$

Note: $J = 2I$, where I = Moment of Inertia.
for circular cross section only.

Torsion of a thin rectangular cross section:

A narrow cross-sectional beam is a rectangular beam that has a width that is much greater than the thickness. Since the cross section is not circular the stress will vary on the outside. As we can see, the stress is max at the center where the stress in a circular cross-section would be generally zero.



Structural Members Under Combined Loading:-

(5)

In many structures the members are required to resist more than 1 kind of loading. These often can be analyzed by superimposing the stresses and strains caused by each load acting separately.

Superposition of stresses is permissible only under the following conditions:-

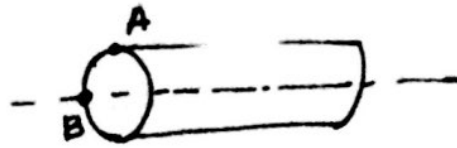
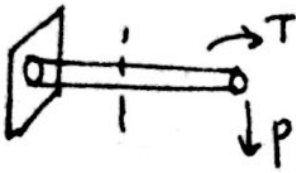
- (a) Stresses must be a linear function of the applied loads. i.e. displacements should be small and Hooke's law must be obeyed.
- (b) There must not be any interaction between the various loads.

Method of Analysis:-

- 1) Select the point on the structure where stresses are to be determined.
 - 2) For each load determine stress resultant at the selected point.
 - 3) Calculate normal & shear stress at the selected point due to the stress resultants.
 - 4) Combine individual stresses to obtain the resultant stresses at the selected point.
 - 5) Select additional points and repeat the process.
- Normal stress due to axial load $\Rightarrow \sigma = P/A$
due to bending $\sigma = -\frac{My}{I}$

The bar shown is subjected to two types of loads.
A torque T and Vertical load P

(6)



Now, let us select two points, one (A) on top of the bar and (B) at the side of the bar. but in the same cross section.

The resulting stress acting across the section are:

- (a) Twisting moment equal to the torque T
- (b) Bending moment M equal to $P \times \text{distance of } B'$
- (c) Shear force equal to P .

The torque T produces stress $\tau_{\max} = \frac{T \cdot R}{J}$

Now this stress due to the torque acts horizontally to point A and Vertical to point B. at the cross section.

The stress due to bending is given by

$$\sigma_{\text{bend}} = \frac{M y}{I} \text{ where } y \text{ can be replaced by } R. \frac{M R}{I}$$

The point to note is that the bending does not produce any moment about B' as it is in the neutral plane.

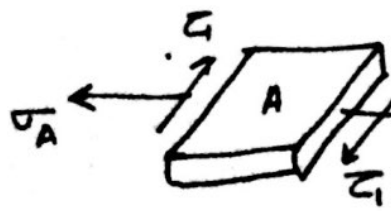
Now, the shear force due to bending will act at B' and will have not affect on A' .

Hence, at point B' it will be an addition of

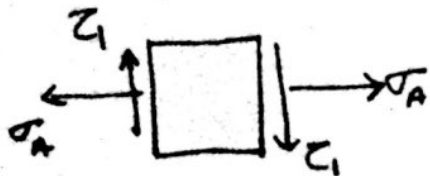
τ_B , shear stress due to bending and τ_T shear stress due to the torque. And at point A'

we have the bending stress and the shear stress due to torque acting.

Now, let us consider the element at point 'A', which would be placed horizontally. We have an axial stress due to bending and a torque shear stress



Redrawing the element into a 2-D element, we get,



From the fig we can see that

$$\sigma_x = \sigma_A \text{ and } \sigma_y = 0$$

and $\tau_{xy} = -\tau_1$. Hence, this becomes a

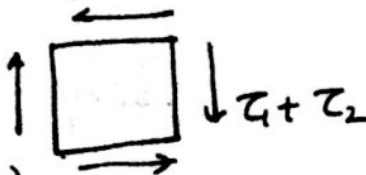
plane stress condition.

Now, consider the element at 'B'.

Hence, drawing the 2-D shape of the element we get,

$$\text{Hence, } \sigma_x = \sigma_y = 0$$

$$\text{and } \tau_{xy} = -(\tau_1 + \tau_2).$$

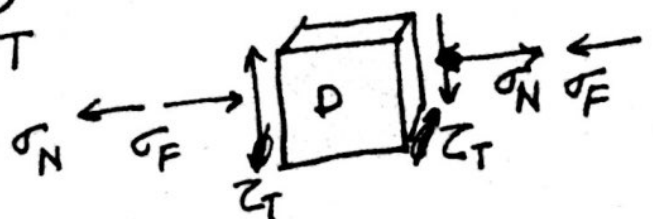
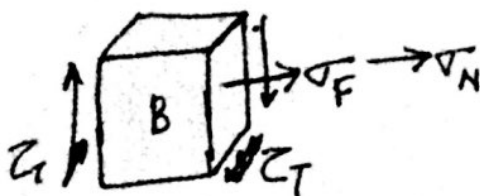
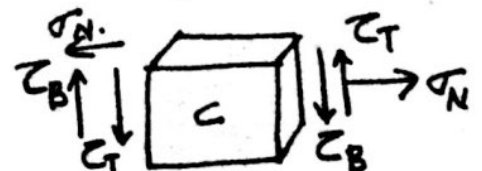
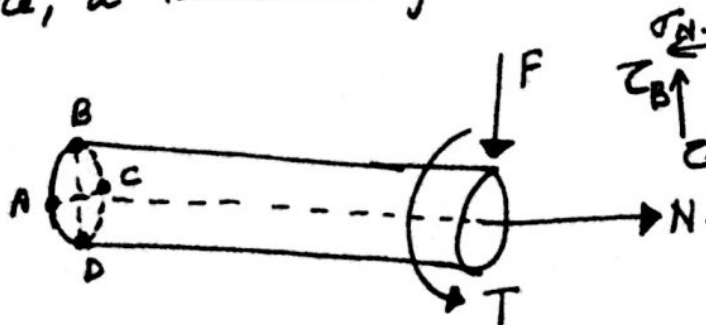
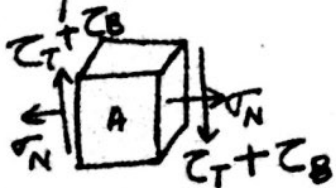


Selection of Critical Areas and points:-

To select the points which have stress resultants at their largest values, it is important to select points where the normal stresses or shear stresses have their largest

values.

Now, let us see the circular bar shown, below subjected to a normal force, a transverse force and a Torque at the free end.



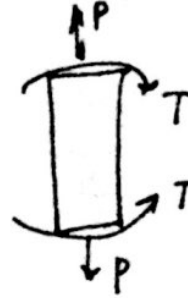
(8)

Example: The rotor of Helicopter drives the rotor, as a consequence, the shaft is subjected to a combination of torsion and axial loading.

$$\phi = 50 \text{ mm}, T = 2.4 \text{ kN}\cdot\text{m}, P = 125 \text{ kN (Tensile)}$$

Determine max tensile stress & maximum comp. stress and max shear stress.

$$\sigma = P/A = \frac{125 \text{ kN}}{\frac{\pi}{4} (0.05)^2} = 63.66 \text{ mpa.}$$



$$\tau = \frac{T \cdot R}{J} = \frac{(2.4 \text{ kN}\cdot\text{m})(0.05/2)}{\frac{\pi}{32} (0.05)^4} = 97.78 \text{ mpa}$$

Hence, now, we know $\sigma_y = 63.66 \text{ mpa}$, $\sigma_x = 0$
 $\tau_{xy} = 97.78 \text{ mpa}$

$$\sigma_{1,2} = \left[\frac{\sigma_x + \sigma_y}{2} \right] \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + (\tau_{xy})^2}$$

$$\sigma_1 = 135 \text{ mpa}$$

$$\sigma_2 = -71 \text{ mpa.}$$

$$\tau_{\max} = \sqrt{\frac{4052.59}{2} + 9560.92}$$

$$\tau_{\max} = 107.64 \text{ mpa.}$$